

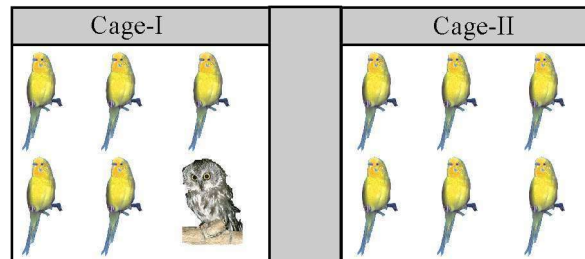
DPP - 6

- Q.1 A bowl has 6 red marbles and 3 green marbles. The probability that a blind folded person will draw a red marble on the second draw from the bowl without replacing the marble from the first draw, is
(A) $\frac{2}{3}$ (B) $\frac{1}{4}$ (C) $\frac{5}{12}$ (D) $\frac{5}{8}$
- Q.2 The probability that a radar will detect an object in one cycle is p . The probability that the object will be detected in n cycles is :
(A) $1 - p^n$ (B) $1 - (1 - p)^n$ (C) p^n (D) $p(1 - p)^{n-1}$
- Q.3 In a certain factory, machines A, B and C produce bolts. Of their production, machines A, B, and C produce 2%, 1% and 3% defective bolts respectively. Machine A produces 35% of the total output of bolts, machine B produces 25% and machine C produces 40%. A bolts is chosen at random from the factory's production and is found to be defective. The probability it was produced on machine C, is
(A) $\frac{6}{11}$ (B) $\frac{23}{45}$ (C) $\frac{24}{43}$ (D) $\frac{3}{11}$
- Q.4 Mr. Dupont is a professional wine taster. When given a French wine, he will identify it with probability 0.9 correctly as French, and will mistake it for a Californian wine with probability 0.1. When given a Californian wine, he will identify it with probability 0.8 correctly as Californian, and will mistake it for a French wine with probability 0.2. Suppose that Mr. Dupont is given ten unlabelled glasses of wine, three with French and seven with Californian wines. He randomly picks a glass, tries the wine, and solemnly says : "French". The probability that the wine he tasted was Californian, is nearly equal to
(A) 0.14 (B) 0.24 (C) 0.34 (D) 0.44
- Q.5 Three numbers are chosen at random without replacement from $\{1, 2, 3, \dots, 10\}$. The probability that the minimum of the chosen numbers is 3 or their maximum is 7 is
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{11}{40}$
- Q.6 Two buses A and B are scheduled to arrive at a town central bus station at noon. The probability that bus A will be late is $\frac{1}{5}$. The probability that bus B will be late is $\frac{7}{25}$. The probability that the bus B is late given that bus A is late is $\frac{9}{10}$. Then the probabilities
(i) neither bus will be late on a particular day and
(ii) bus A is late given that bus B is late, are respectively
(A) $\frac{2}{25}$ and $\frac{12}{28}$ (B) $\frac{18}{25}$ and $\frac{22}{28}$ (C) $\frac{7}{10}$ and $\frac{18}{28}$ (D) $\frac{12}{25}$ and $\frac{2}{28}$
- Q.7 If at least one child in a family with 3 children is a boy then the probability that exactly 2 of the children are boys, is
(A) $\frac{3}{7}$ (B) $\frac{4}{7}$ (C) $\frac{1}{3}$ (D) $\frac{3}{8}$
- Q.8 From an urn containing six balls, 3 white and 3 black ones, a person selects at random an even number of balls (all the different ways of drawing an even number of balls are considered equally probable, irrespective of their number). Then the probability that there will be the same number of black and white balls among them
(A) $\frac{4}{5}$ (B) $\frac{11}{15}$ (C) $\frac{11}{30}$ (D) $\frac{2}{5}$

- Q.9 There are three main political parties namely 1, 2, 3. If in the adjoining table p_{ij} , ($i, j=1, 2, 3$) denote the probability that party j wins the general elections contested when party i is in the power. What is the probability that the party 2 will be in power after the next two elections, given that the party 1 is in the power?

$P_{11}=0.7$	$P_{12}=0.2$	$P_{13}=0.1$
$P_{21}=0.5$	$P_{22}=0.3$	$P_{23}=0.2$
$P_{31}=0.3$	$P_{32}=0.4$	$P_{33}=0.3$

- (A) 0.27 (B) 0.24 (C) 0.14 (D) 0.06
- Q.10 Shalu bought two cages of birds : Cage-I contains 5 parrots and 1 owl, and Cage-II contains 6 parrots, as shown



One day Shalu forgot to lock both cages and two birds flew from Cage-I to Cage-II. Then two birds flew back from Cage-II to Cage-I. Assume that all birds have equal chance of flying, the probability that the Owl is still in Cage-I, is

- (A) $1/6$ (B) $1/3$ (C) $2/3$ (D) $3/4$

[REASONING TYPE]

- Q.11 From a well shuffled pack of 52 playing cards a card is drawn at random. Two events A and B are defined as

A: Red card is drawn.

B: Card drawn is either a Diamond or Heart

Statement-1: $P(A + B) = P(AB)$

because

Statement-2: $A \subseteq B$ and $B \subseteq A$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

Paragraph for question nos. 14 to 16

A box contains b red balls, ' $2b$ ' white balls and ' $3b$ ' blue balls where b is a positive integer. 3 balls are selected at random from the box.

- Q.12 If balls are drawn without replacement and 'A' denotes the event that "No two of the selected balls have the same colour" then

- (A) there is no value of b for which $P(A) = 0.3$
 (B) There is exactly one value of b for which $P(A) = 0.3$ and this value is less than 10.
 (C) There is exactly one value of b for which $P(A) = 0.3$ and this value is greater than 10.
 (D) There is more than one value of b for which $P(A) = 0.3$

- Q.13 If balls are drawn without replacement and 'B' denotes the event that "No two of the 3 drawn balls are blue" then

- (A) $P(B) = \frac{1}{3}$ if $b=1$ (B) $P(B) = \frac{2}{3}$ if $b=2$ (C) $P(B) = \frac{1}{4}$ if $b=4$ (D) $P(B) = \frac{1}{2}$ for all value of b .

- Q.14 If $P(A) = 0.3$, then the value of $P(A/B)$ equals
 (A) $3/5$ (B) $3/10$ (C) $1/2$ (D) $2/3$

Pagagraph for question nos. 15 to 17

Urn-I contains 5 Red balls and 1 Blue ball,

Urn-II contains 2 Red balls and 4 Blue balls.

A fair die is tossed. If it results in an even number, balls are repeatedly withdrawn one at a time with replacement from urn-I. If it is an odd number, balls are repeatedly withdrawn one at a time with replacement from urn-II. Given that the first two draws both have resulted in a blue ball.

- Q.15 Conditional probability that the first two draws have resulted in blue balls given urn-II is used is
 (A) $1/2$ (B) $4/9$ (C) $1/3$ (D) None
- Q.16 If the probability that the urn-I is being used is p , and q is the corresponding figure for urn-II then
 (A) $q = 16p$ (B) $q = 4p$ (C) $q = 2p$ (D) $q = 3p$
- Q.17 The probability of getting a red ball in the third draw, is
 (A) $1/3$ (B) $1/2$ (C) $37/102$ (D) $41/102$

[MULTIPLE OBJECTIVE TYPE]

- Q.18 Two whole numbers are randomly selected and multiplied. Consider two events E_1 and E_2 defined as
 E_1 : Their product is divisible by 5
 E_2 : Unit's place in their product is 5.
 Which of the following statement(s) is/are correct?
 (A) E_1 is twice as likely to occur as E_2 . (B) E_1 and E_2 are disjoint
 (C) $P(E_2/E_1) = 1/4$ (D) $P(E_1/E_2) = 1$

[MATCH THE COLUMN]

- | Q.19 | Column-I | Column-II |
|------|---|----------------|
| (A) | The probability of a bomb hitting a bridge is $1/2$. Two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9, is | (P) 4 |
| (B) | A bag contains 2 red, 3 white and 5 black balls, a ball is drawn its colour is noted and replaced. Minimum number of times, a ball must be drawn so that the probability of getting a red ball for the first time is at least even, is | (Q) 5 |
| (C) | A hunter knows that a deer is hidden in one of the two near by bushes, the probability of its being hidden in bush-I being $4/5$. The hunter having a rifle containing 10 bullets decides to fire them all at bush-I or II. It is known that each shot may hit one of the two bushes, independently of the other with probability $1/2$. Number of bullets must he fire on bush-I to hit the animal with maximum probability is (Assume that the bullet hitting the bush also hits the animal). | (R) 6
(S) 7 |

[SUBJECTIVE]

- Q.20 A lot contains 50 defective & 50 non defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as :
 $A = \{ \text{the first bulb is defective} \}$; $B = \{ \text{the second bulb is non defective} \}$
 $C = \{ \text{the two bulbs are both defective or both non defective} \}$
 Determine whether (i) A,B,C are pair wise independent (ii) A,B,C are independent